HW2: Transfer Learning for Flowers102 Classification

Student Report
Deep Learning and Spatial Intelligence

Abstract

This report presents four experiments on Flowers102 image classification: ImageNet-pretrained ResNet fine-tuning, hyperparameter analysis, pretraining ablation, and attention/Transformer comparison. The baseline ResNet-18 and ResNet-34 models reached 89.85% and 89.64% test accuracy, respectively. Hyperparameter tuning improved the best ResNet result to 91.59% test accuracy with ResNet-34 using classifier learning rate 5×10^{-4} and backbone learning rate 5×10^{-5}. The pretraining ablation showed the largest effect: ResNet-18 dropped from 89.85% to 28.41% test accuracy when trained from scratch, and ResNet-34 dropped from 89.64% to 21.01%. In the final comparison, Swin-T achieved the best overall result with 94.51% best validation accuracy and 92.24% test accuracy.

1 Dataset and Common Setup

The experiments use the 102 Category Flower Dataset through `torchvision.datasets.Flowers102` [Nilsback and Zisserman, 2008]. The official split contains 1020 training images, 1020 validation images, and 6149 test images over 102 classes. This means that each class has only a small number of labeled training examples, so the task is a good setting for studying transfer learning.

For training images, I use `RandomResizedCrop(224)`, `RandomHorizontalFlip`, `ColorJitter`, `ToTensor`, and ImageNet normalization. For validation and test images, I use `Resize(256)`, `CenterCrop(224)`, `ToTensor`, and the same ImageNet normalization. All reported runs use top-1 accuracy as the main metric. The optimizer is AdamW with weight decay 1×10^{-4}, random seed 42, and the MPS device on a local Mac.

2 Task 1: ImageNet-pretrained ResNet Baseline

2.1 Method

For the first task, I fine-tune ResNet-18 and ResNet-34 [He et al., 2016]. Both models are initialized with ImageNet-pretrained weights [Deng et al., 2009]. I replace the original 1000-way fully connected layer with a new 102-way classifier for Flowers102. The pretrained backbone and the new classifier use different learning rates: the backbone learning rate is 1×10^{-4}, while the classifier learning rate is 1×10^{-3}. This design lets the ImageNet features change slowly while allowing the randomly initialized classifier to adapt quickly.

2.2 Results

Table 1 shows the baseline results. ResNet-18 reached 92.16% best validation accuracy and 89.85% test accuracy. ResNet-34 reached 91.76% best validation accuracy and 89.64% test accuracy. In this baseline setting, the deeper ResNet-34 did not outperform ResNet-18; the two models are very close, with ResNet-18 slightly higher by 0.21 percentage points on the test split.

Table 1: Task 1 baseline fine-tuning results. Accuracies are percentages.

Folder	Model	Pretrained	Epochs	BS	Backbone LR	Classifier LR	Best Val	Test
task1_resnet18	resnet18	True	30	32	1e-4	1e-3	92.16	89.85
task1_resnet34	resnet34	True	30	32	1e-4	1e-3	91.76	89.64

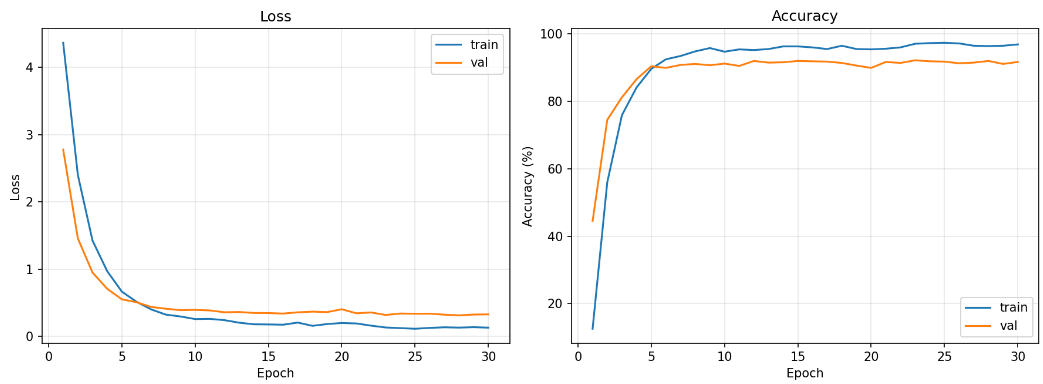


Figure 1: Task 1 ResNet-18 baseline training and validation curves.

The ResNet-18 training curve in Figure 1 shows fast improvement in the first few epochs. The best validation checkpoint appears before the last epoch, so saving the best validation model is important for this dataset.

3 Task 2: Hyperparameter Analysis

3.1 Method

For Task 2, I keep ImageNet-pretrained ResNet-18 and ResNet-34, but manually change epoch count, classifier learning rate, backbone learning rate, and batch size. The classifier learning rate is treated as the main learning rate, while the backbone learning rate is set to one tenth of it. The tested classifier learning rates are 1×10^{-4}, 5×10^{-4}, 1×10^{-3}, and 2×10^{-3}. I also compare 10, 30, and 50 training epochs, and batch sizes 16 and 32.

3.2 Results for ResNet-18

Table 2 gives the ResNet-18 hyperparameter results. The best ResNet-18 setting is 30 epochs, batch size 32, backbone learning rate 5×10^{-5}, and classifier learning rate 5×10^{-4}, which reaches 94.12% best validation accuracy and 90.62% test accuracy. Compared with the Task 1 baseline test accuracy of 89.85%, this is a 0.77 point improvement.

The learning rate comparison is the clearest part of the result. With classifier learning rate 1×10^{-4}, ResNet-18 only reaches 85.59% test accuracy, which suggests slow adaptation. With classifier learning rate 2×10^{-3}, test accuracy is 85.64%, suggesting the update is too aggressive. The moderate 5×10^{-4} setting performs best. Training for 50 epochs does not improve test accuracy over 30 epochs under the standard 1×10^{-3} classifier learning rate, and reducing batch size from 32 to 16 lowers test accuracy from 89.85% to 87.40%.

3.3 Results for ResNet-34

Table 3 gives the ResNet-34 hyperparameter results. The best ResNet-34 setting is also classifier learning rate 5×10^{-4} and backbone learning rate 5×10^{-5}, reaching 93.53% best validation accuracy and 91.59% test accuracy. This is the best ResNet-only result in the whole experiment.

The same pattern appears for ResNet-34: 1×10^{-4} is too small and gives 88.06% test accuracy, while 2×10^{-3} is too large and gives 85.85%. The tuned 5×10^{-4} setting is clearly better than the

Table 2: Task 2 hyperparameter analysis for ResNet-18.

Epochs	BS	Backbone LR	Classifier LR	Best Val	Test
10	32	1e-4	1e-3	91.18	89.19
30	16	1e-4	1e-3	91.76	87.40
30	32	1e-5	1e-4	89.22	85.59
30	32	5e-5	5e-4	94.12	90.62
30	32	1e-4	1e-3	92.16	89.85
30	32	2e-4	2e-3	89.31	85.64
50	32	1e-4	1e-3	92.25	89.74

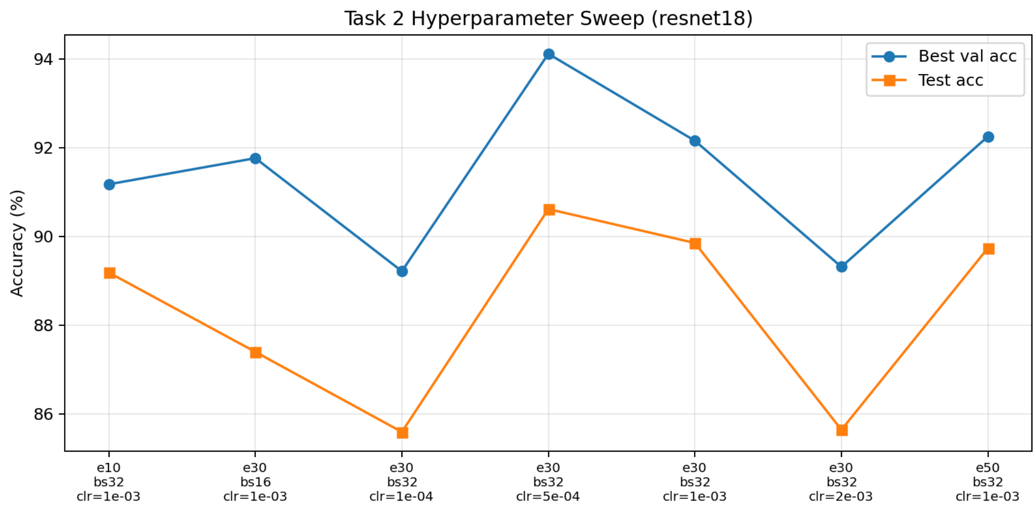


Figure 2: Task 2 ResNet-18 hyperparameter sweep. Each tick shows epochs, batch size, and classifier learning rate.

original 1×10^{-3} baseline. Under the original learning rate, increasing epochs from 30 to 50 does not improve the test result, and the batch size 16 run is lower than batch size 32. These results suggest that learning rate tuning matters more than simply training longer.

4 Task 3: Pretraining Ablation

4.1 Method

For Task 3, I compare ImageNet-pretrained initialization with random initialization. The network architecture is still ResNet-18 or ResNet-34, and the output layer is changed to 102 classes. The pretrained runs use backbone learning rate 1×10^{-4} and classifier learning rate 1×10^{-3}. The scratch runs use learning rate 1×10^{-3} for both backbone and classifier, since all parameters must be learned from the Flowers102 training set.

4.2 Results

Table 4 shows that pretraining is the most important factor in the project. ResNet-18 improves from 28.41% test accuracy when trained from scratch to 89.85% with ImageNet pretraining, a gain of 61.44 percentage points. ResNet-34 improves from 21.01% to 89.64%, a gain of 68.63 percentage points.

The curves in Figure 4 make this difference visible. The pretrained ResNet-18 quickly reaches high validation accuracy, while the scratch model remains around 30% validation accuracy after 30 epochs. This supports the conclusion that Flowers102 is too small for training these ResNets from random initialization within the current training budget.

Table 3: Task 2 hyperparameter analysis for ResNet-34.

Epochs	BS	Backbone LR	Classifier LR	Best Val	Test
10	32	1e-4	1e-3	91.76	89.64
30	16	1e-4	1e-3	91.18	88.78
30	32	1e-5	1e-4	90.98	88.06
30	32	5e-5	5e-4	93.53	91.59
30	32	1e-4	1e-3	91.76	89.64
30	32	2e-4	2e-3	87.84	85.85
50	32	1e-4	1e-3	91.76	89.64

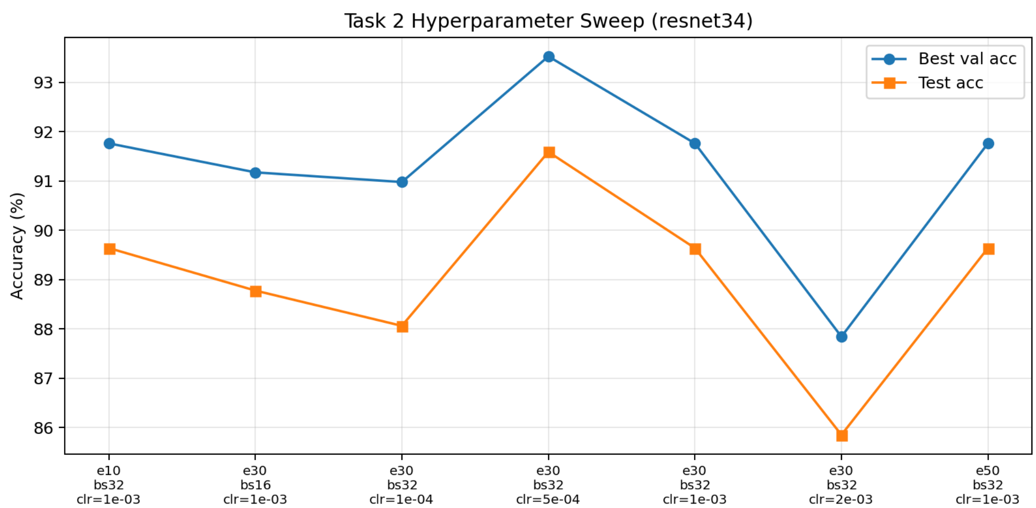


Figure 3: Task 2 ResNet-34 hyperparameter sweep. The 5×10^{-4} classifier learning rate gives the best ResNet-only test accuracy.

5 Task 4: Attention Mechanisms and Transformer Models

5.1 Method

For Task 4, I test two directions. First, I manually add SE-block and CBAM to the ResNet BasicBlock. SE-block applies channel-wise attention using global pooling and a small gating network [Hu et al., 2018]. CBAM applies both channel attention and spatial attention [Woo et al., 2018]. The attention ResNets reuse matching ResNet pretrained weights when possible, while the new attention modules are randomly initialized.

Second, I compare two lightweight Transformer models: ViT-Tiny and Swin-T [Dosovitskiy et al., 2021, Liu et al., 2021]. Both are initialized with pretrained weights and use a 102-class output head. ViT-Tiny uses batch size 8 and learning rate 1×10^{-4}, while the recorded Swin-T run uses batch size 32 and learning rate 1×10^{-4}.

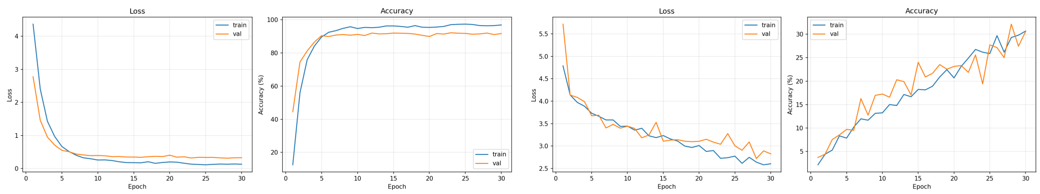
5.2 Results

Table 5 summarizes the attention and Transformer results. In these runs, SE and CBAM do not improve over the plain ResNet baselines. For ResNet-18, SE reaches 87.87% test accuracy and CBAM reaches 81.36%, both below the 89.85% baseline. For ResNet-34, SE reaches 87.23% and CBAM reaches 83.77%, both below the 89.64% baseline.

SE is more stable than CBAM in both depths, but still lower than the plain baselines. A likely reason is that the newly inserted attention layers are randomly initialized, and the small training split may not be enough to tune them well without additional hyperparameter search. Figure 5 shows the same

Table 4: Task 3 pretraining ablation.

Model	Initialization	Best Val	Test
resnet18	pretrained	92.16	89.85
resnet18	scratch	32.16	28.41
resnet34	pretrained	91.76	89.64
resnet34	scratch	25.98	21.01



Pretrained Scratch

Figure 4: Task 3 ResNet-18 pretraining ablation curves.

conclusion visually. The SE-ResNet-18 curve in Figure 6 shows successful training, but its final generalization is still weaker than the plain ResNet-18 run.

For Transformer models, ViT-Tiny reaches 87.58% test accuracy, which is lower than the ResNet baselines. Swin-T reaches 94.51% best validation accuracy and 92.24% test accuracy, making it the best overall model in my experiments. I do not interpret the ViT-Tiny result as evidence that ViT is generally worse; it may need a more suitable fine-tuning recipe. However, the Swin-T result shows that a pretrained hierarchical Transformer can be very competitive on Flowers102.

6 Discussion

Across the four tasks, ImageNet pretraining is the strongest and most reliable factor. The scratch models perform poorly because the Flowers102 training set contains only 1020 images, so random initialization has too little supervision to learn robust visual representations. Hyperparameter tuning also matters: the best ResNet runs use a slightly smaller classifier learning rate than the baseline. Training longer does not automatically improve test accuracy, which suggests that overfitting and checkpoint selection are important.

The comparison between ResNet-18 and ResNet-34 is not one-sided. ResNet-18 is slightly better in the baseline, while ResNet-34 becomes better after learning-rate tuning. Attention modules are not stable improvements here, and CBAM hurts more than SE. Among all models, Swin-T gives the highest test accuracy, but this conclusion is based on the tested hyperparameters only.

7 Conclusion

The best overall model is Swin-T with 92.24% test accuracy. The best ResNet-only model is ResNet-34 with classifier learning rate 5×10^{-4}, backbone learning rate 5×10^{-5}, batch size 32, and 30 epochs, reaching 91.59% test accuracy. ImageNet pretraining is essential: it improves ResNet-18 by 61.44 percentage points and ResNet-34 by 68.63 percentage points over scratch training. The main limitation is that the experiments use a limited hyperparameter grid and were run on local Mac hardware, so future work should include repeated seeds, learning-rate schedules, and more architecture-specific tuning.

A Complete Results

Table 6 lists all experiment folders used in this report.

Table 5: Task 4 attention and Transformer comparison.

Model	Attention / Architecture	Pretrained	Best Val	Test
resnet18	se	True	90.78	87.87
resnet18	cbam	True	85.29	81.36
resnet34	se	True	90.20	87.23
resnet34	cbam	True	88.24	83.77
vit_tiny	vit_tiny	True	91.27	87.58
swin_t	swin_t	True	94.51	92.24

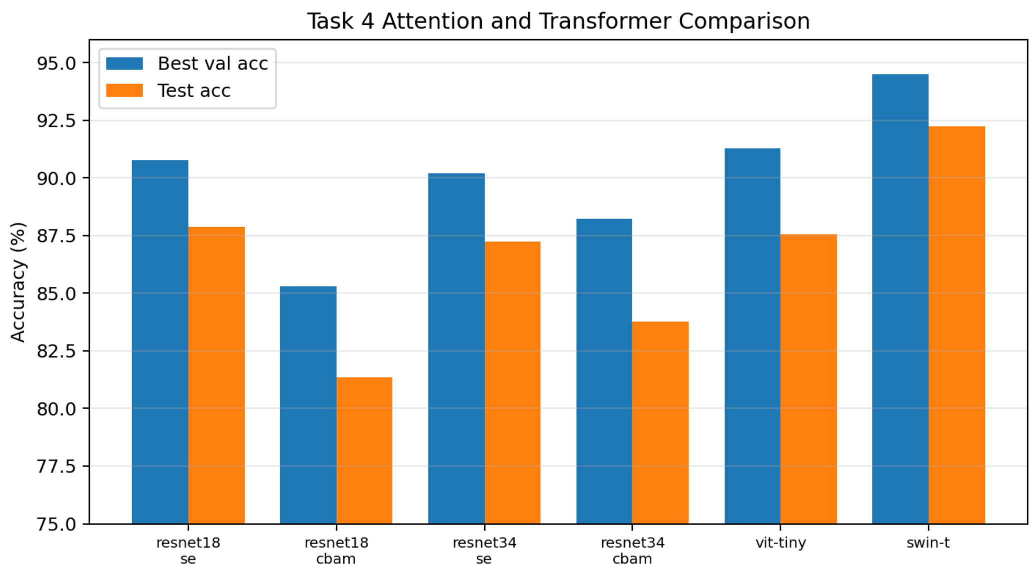


Figure 5: Task 4 comparison of attention ResNets and Transformer models. Swin-T is the strongest model in this experiment.

B Representative Commands

```
python scripts/prepare_data.py --data_dir ./data
python scripts/train_task1_baseline.py --model resnet18 --data_dir ./data \
  --epochs 30 --batch_size 32 --backbone_lr 1e-4 --classifier_lr 1e-3 \
  --weight_decay 1e-4 --optimizer adamw --num_workers 0 --device mps
python scripts/train_task4_attention.py --model swin_t --pretrained true \
  --data_dir ./data --epochs 30 --batch_size 32 --lr 1e-4 \
  --weight_decay 1e-4 --optimizer adamw --num_workers 0 --device mps
```

References

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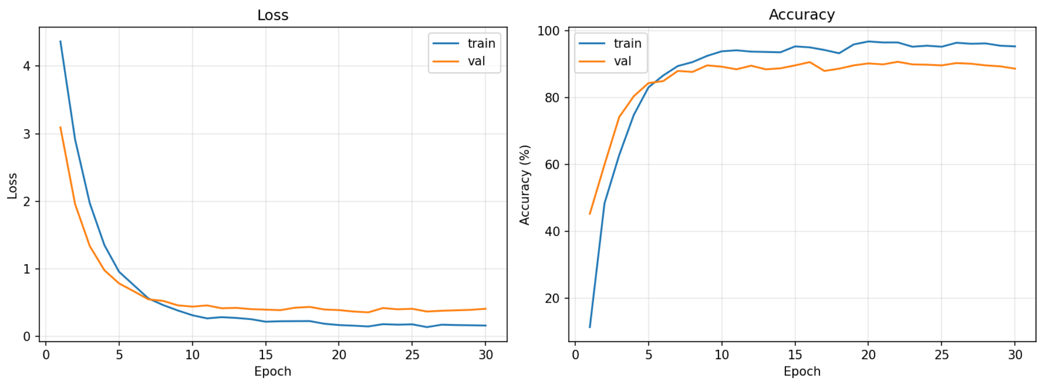

Figure 6: Task 4 SE-ResNet-18 training and validation curves.

Table 6: Complete experiment summary.

Task	Model	Attn	Init	Epochs	BS	BB LR	Cls LR	LR	Best Val	Test	Folder
task1_baseline	resnet18	none	pretrained	30	32	1e-4	1e-3	–	92.16	89.85	task1_resnet18
task1_baseline	resnet34	none	pretrained	30	32	1e-4	1e-3	–	91.76	89.64	task1_resnet34
task2_hparam	resnet18	none	pretrained	10	32	1e-4	1e-3	–	91.18	89.19	task2_resnet18_epoch10_blr1e-4_clr1e-3_bs32
task2_hparam	resnet18	none	pretrained	30	16	1e-4	1e-3	–	91.76	87.40	task2_resnet18_epoch30_blr1e-4_clr1e-3_bs16
task2_hparam	resnet18	none	pretrained	30	32	1e-5	1e-4	–	89.22	85.59	task2_resnet18_epoch30_blr1e-5_clr1e-4_bs32
task2_hparam	resnet18	none	pretrained	30	32	5e-5	5e-4	–	94.12	90.62	task2_resnet18_epoch30_blr5e-5_clr5e-4_bs32
task2_hparam	resnet18	none	pretrained	30	32	1e-4	1e-3	–	92.16	89.85	task2_resnet18_epoch30_blr1e-4_clr1e-3_bs32
task2_hparam	resnet18	none	pretrained	30	32	2e-4	2e-3	–	89.31	85.64	task2_resnet18_epoch30_blr2e-4_clr2e-3_bs32
task2_hparam	resnet18	none	pretrained	50	32	1e-4	1e-3	–	92.25	89.74	task2_resnet18_epoch50_blr1e-4_clr1e-3_bs32
task2_hparam	resnet34	none	pretrained	10	32	1e-4	1e-3	–	91.76	89.64	task2_resnet34_epoch10_blr1e-4_clr1e-3_bs32
task2_hparam	resnet34	none	pretrained	30	16	1e-4	1e-3	–	91.18	88.78	task2_resnet34_epoch30_blr1e-4_clr1e-3_bs16
task2_hparam	resnet34	none	pretrained	30	32	1e-5	1e-4	–	90.98	88.06	task2_resnet34_epoch30_blr1e-5_clr1e-4_bs32
task2_hparam	resnet34	none	pretrained	30	32	5e-5	5e-4	–	93.53	91.59	task2_resnet34_epoch30_blr5e-5_clr5e-4_bs32
task2_hparam	resnet34	none	pretrained	30	32	1e-4	1e-3	–	91.76	89.64	task2_resnet34_epoch30_blr1e-4_clr1e-3_bs32
task2_hparam	resnet34	none	pretrained	30	32	2e-4	2e-3	–	87.84	85.85	task2_resnet34_epoch30_blr2e-4_clr2e-3_bs32
task2_hparam	resnet34	none	pretrained	50	32	1e-4	1e-3	–	91.76	89.64	task2_resnet34_epoch50_blr1e-4_clr1e-3_bs32
task3_ablation	resnet18	none	pretrained	30	32	1e-4	1e-3	–	92.16	89.85	task3_resnet18_pretrained
task3_ablation	resnet18	none	scratch	30	32	1e-3	1e-3	–	32.16	28.41	task3_resnet18_scratch
task3_ablation	resnet34	none	pretrained	30	32	1e-4	1e-3	–	91.76	89.64	task3_resnet34_pretrained
task3_ablation	resnet34	none	scratch	30	32	1e-3	1e-3	–	25.98	21.01	task3_resnet34_scratch
task4_attention	resnet18	cbam	pretrained	30	32	1e-4	1e-3	1e-4	85.29	81.36	task4_resnet18_cbam
task4_attention	resnet18	se	pretrained	30	32	1e-4	1e-3	1e-4	90.78	87.87	task4_resnet18_se
task4_attention	resnet34	cbam	pretrained	30	32	1e-4	1e-3	1e-4	88.24	83.77	task4_resnet34_cbam
task4_attention	resnet34	se	pretrained	30	32	1e-4	1e-3	1e-4	90.20	87.23	task4_resnet34_se
task4_attention	vit_tiny	none	pretrained	30	8	1e-4	1e-3	1e-4	91.27	87.58	task4_vit_tiny
task4_attention	swin_t	none	pretrained	30	32	1e-4	1e-3	1e-4	94.51	92.24	task4_swin_t

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